

Primero actualizamos los repositorios:

```
ubuntu@ip-172-31-25-6:~$ sudo apt update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe Trans
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64
Get:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 Package
Get:9 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64
Get:10 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse am
Get:11 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse Tr
Get:12 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse am
Get:13 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse am
Get:14 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main
Get:15 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main
Get:16 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main
```

Instalamos algunos paquetes de requisitos previos que permitan a apt usar paquetes a través de HTTPS:

```
ubuntu@ip-172-31-25-6:~$ sudo apt install apt-transport-https ca-certificates curl software-properties-common
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ca-certificates is already the newest version (20240203).
ca-certificates set to manually installed.
curl is already the newest version (8.5.0-2ubuntu10.6).
curl set to manually installed.
software-properties-common is already the newest version (0.99.49.3).
software-properties-common set to manually installed.
The following NEW packages will be installed:
  apt-transport-https
0 upgraded, 1 newly installed, 0 to remove and 38 not upgraded.
Need to get 3970 B of archives.
After this operation, 36.9 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 apt-transport-https all 2.8.
```

Añadimos la clave de GPG para el repositorio oficial de Docker en el sistema y después agregamos el repositorio de Docker a las fuentes de APT:

```
ubuntu@ip-172-31-25-6:~$ curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo apt-key add -
Warning: apt-key is deprecated. Manage keyring files in trusted.gpg.d instead (see apt-key(8)).
OK
ubuntu@ip-172-31-25-6:~$ sudo add-apt-repository "deb [arch=amd64] https://download.docker.com/linux/ubuntu focal stable"
Repository: 'deb [arch=amd64] https://download.docker.com/linux/ubuntu focal stable'
Description:
Archive for codename: focal components: stable
More info: https://download.docker.com/linux/ubuntu
Adding repository.
Press [ENTER] to continue or Ctrl-c to cancel.
Adding deb entry to /etc/apt/sources.list.d/archive_uri-https_download_docker_com_linux_ubuntu-noble.l
Adding disabled deb-src entry to /etc/apt/sources.list.d/archive_uri-https_download_docker_com_linux_u
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Get:5 https://download.docker.com/linux/ubuntu focal InRelease [57.7 kB]
Get:6 https://download.docker.com/linux/ubuntu focal/stable amd64 Packages [59.9 kB]
Fetched 118 kB in 0s (244 kB/s)
Reading package lists... Done
W: https://download.docker.com/linux/ubuntu/dists/focal/InRelease: Key is stored in legacy trusted.gpg
ubuntu@ip-172-31-25-6:~$
```

Actualizamos el paquete de base de datos con los paquetes de Docker del repositorio recién agregado:

```
ubuntu@ip-172-31-25-6:~$ sudo apt update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu InRelease
Hit:4 https://download.docker.com/linux/ubuntu InRelease
Hit:5 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
38 packages can be upgraded. Run 'apt list --upgradable' to see them.
W: https://download.docker.com/linux/ubuntu InRelease
ubuntu@ip-172-31-25-6:~$
```

Instalamos docker

```
ubuntu@ip-172-31-25-6:~$ sudo apt install docker-ce
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  containerd.io docker-buildx-plugin docker-ce-cli docker-ce
Suggested packages:
  cgroupfs-mount | cgroup-lite
The following NEW packages will be installed:
  containerd.io docker-buildx-plugin docker-ce docker-ce-cli
0 upgraded, 10 newly installed, 0 to remove and 38 not upgraded
Need to get 120 MB of archives.
After this operation, 440 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
```

Comprobamos que funciona:

```
ubuntu@ip-172-31-25-6:~$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: active (running) since Fri 2025-09-26 11:17:25 UTC; 1min 10s ago
     TriggeredBy: ● docker.socket
   Docs: https://docs.docker.com
  Main PID: 3386 (dockerd)
    Tasks: 9
   Memory: 44.7M (peak: 44.9M)
      CPU: 373ms
   CGroup: /system.slice/docker.service
           └─3386 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.s
```

Por defecto, el comando docker solo puede ser ejecutado por el usuario **root** o un usuario del **grupo docker**, que se crea automáticamente durante el proceso de instalación de Docker. Así que nos ponemos los permisos para ejecutarlo:

```
ubuntu@ip-172-31-25-6:~$ sudo usermod -aG docker ${USER}
```

Ahora apagamos y encendemos el servidor para que se actualicen los permisos:

Una vez volvemos a entrar comprobamos que tenemos docker funcionando

```
ubuntu@ip-172-31-25-6:~$ service docker status
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: active (running) since Fri 2025-09-26 11:26:20 UTC; 5min ago
     TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
    Main PID: 746 (dockerd)
      Tasks: 10
     Memory: 91.9M (peak: 92.8M)
        CPU: 670ms
      CGroup: /system.slice/docker.service
             └─746 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock
```

Instalamos apache2

```
ubuntu@ip-172-31-25-6:~$ sudo apt-get install apache2
```

Comprobamos que está instalando

```
ubuntu@ip-172-31-25-6:~$ service apache2 status
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Fri 2025-09-26 11:33:41 UTC; 58s ago
     Docs: https://httpd.apache.org/docs/2.4/
    Main PID: 1890 (apache2)
      Tasks: 55 (limit: 1017)
     Memory: 5.3M (peak: 5.8M)
        CPU: 42ms
      CGroup: /system.slice/apache2.service
             └─1890 /usr/sbin/apache2 -k start
               └─1892 /usr/sbin/apache2 -k start
                 └─1893 /usr/sbin/apache2 -k start

Sep 26 11:33:41 ip-172-31-25-6 systemd[1]: Starting apache2.service - The Apache HTTP Server.
Sep 26 11:33:41 ip-172-31-25-6 systemd[1]: Started apache2.service - The Apache HTTP Server.
```

Creamos un repositorio de docker

```
ubuntu@ip-172-31-25-6:~$ docker run -it --name Web -p 8080:80 ubuntu:22.04 /bin/bash
root@fb86a7697a02:/#
```

Una vez dentro del docker actualizamos e instalamos apache

```
root@fb86a7697a02:/# apt-get install apache2
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
E: Unable to locate package apache2
root@fb86a7697a02:/# apt update
Get:1 http://security.ubuntu.com/ubuntu jammy-s
Get:2 http://archive.ubuntu.com/ubuntu jammy In
```

```
root@fb86a7697a02:/# apt install apache2
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils bzip2 ca-certificates file liba
  libldap-2.5-0 libldap-common liblua5.3-0 libmagic-mgc libmagic1 libnght
  mime-support netbase openssl perl perl-modules-5.34 publicsuffix ssl-c
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-browser
  libsasl2-modules-sql perl-doc libterm-readline-gnu-perl | libterm-read
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils bzip2 ca-certificates f
```

Añadimos la regla de HTTP con puerta 80 en nuestra instancia de AWS:

Entramos en el grupo de seguridad de nuestra instancia y añadimos la regla

Grupos de seguridad

sg-0cb2a78562db8f184 - launch-wizard-2

Acciones

Detalle

Nombre del grupo de seguridad

launch-wizard-2

ID del grupo de seguridad

sg-0cb2a78562db8f184

Descripción

launch-wizard-2 created 2025-09-26T10:58:53.766Z

ID de la VPC

vpc-01a6278e43364aba2

Propietario

381491894262

Número de reglas de entrada

2 Entradas de permisos

Número de reglas de salida

1 Entrada de permiso

Reglas de entrada

Reglas de salida

Compartiendo : novedad

Asociaciones de VPC : novedad

Etiquetas

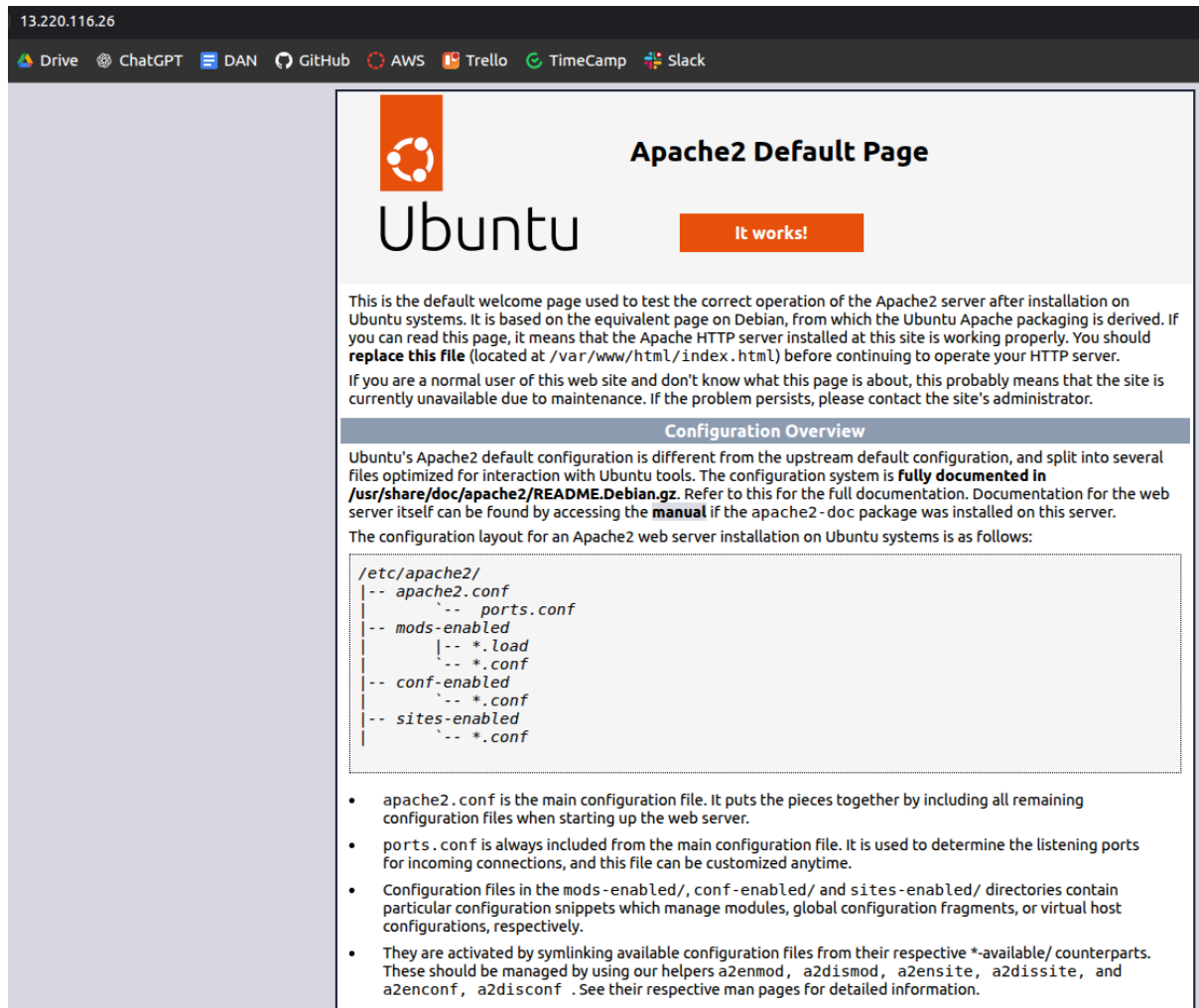
Reglas de entrada (2)

Administrar etiquetas

Editar reglas de entrada

	Name	ID de la regla del gr...	Versión de IP	Tipo	Protocolo	Intervalo de puertos	Origen	Descripción
☐	-	sgr-Ode15b9eab310eaf6	IPv4	SSH	TCP	22	0.0.0.0/0	-
☐	-	sgr-Od37130a4ad957aaa	IPv4	HTTP	TCP	80	0.0.0.0/0	-

Visualizamos apache poniendo la ip publica



Ahora instalamos en el servidor lynx para visualizar el contenido de la web desde el servidor

```
ubuntu@ip-172-31-25-6:~$ sudo apt-get install lynx -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  bzip2 lynx-common mailcap
Suggested packages:
  bzip2-doc
The following NEW packages will be installed:
  bzip2 lynx lynx-common mailcap
0 upgraded, 4 newly installed, 0 to remove and 38 not upg
Need to get 1779 kB of archives.
After this operation, 5937 kB of additional disk space wi
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu nobl
0.1 [34.5 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu nobl
```

Ahora usamos el lynx con la ip pública de la instancia para visualizar apache


```
ubuntu@ip-172-31-25-6:~$ lynx http://13.220.116.26/
```

```

Apache2 Ubuntu Default Page: It works (p1 of 2)

Ubuntu Logo
Apache2 Default Page
It works!

This is the default welcome page used to test the correct operation of the Apache2 server after
installation on Ubuntu systems. It is based on the equivalent page on Debian, from which the
Ubuntu Apache packaging is derived. If you can read this page, it means that the Apache HTTP
server installed at this site is working properly. You should replace this file (located at
/var/www/html/index.html) before continuing to operate your HTTP server.

If you are a normal user of this web site and don't know what this page is about, this probably
means that the site is currently unavailable due to maintenance. If the problem persists,
please contact the site's administrator.
Configuration Overview

Ubuntu's Apache2 default configuration is different from the upstream default configuration,
and split into several files optimized for interaction with Ubuntu tools. The configuration
system is fully documented in /usr/share/doc/apache2/README.Debian.gz. Refer to this for the
full documentation. Documentation for the web server itself can be found by accessing the
manual if the apache2-doc package was installed on this server.

The configuration layout for an Apache2 web server installation on Ubuntu systems is as
follows:
etc/apache2/
-- apache2.conf
    |-- ports.conf
-- mods-enabled
    |-- *.load
    |-- *.conf
-- conf-enabled
    |-- *.conf
-- sites-enabled
    |-- *.conf

* apache2.conf is the main configuration file. It puts the pieces together by including all
remaining configuration files when starting up the web server.
* ports.conf is always included from the main configuration file. It is used to determine the
listening ports for incoming connections, and this file can be customized anytime.
* Configuration files in the mods-enabled/, conf-enabled/ and sites-enabled/ directories
contain particular configuration snippets which manage modules, global configuration
fragments, or virtual host configurations, respectively.
* They are activated by symlinking available configuration files from their respective
*-available/ counterparts. These should be managed by using our helpers a2enmod, a2dismod,
a2ensite, a2dissite, and a2enconf, a2disconf . See their respective man pages for detailed
information.
* The binary is called apache2 and is managed using systemd, so to start/stop the service use
systemctl start apache2 and systemctl stop apache2, and use systemctl status apache2 and
- press space for next page -
Arrow keys: Up and Down to move. Right to follow a link; Left to go back.
H)elp O)ptions P)rint G)o M)ain screen Q)uit /=search [delete]=history list
```

docker ps -a

```
ubuntu@ip-172-31-25-6:~$ docker ps -a
CONTAINER ID   IMAGE      COMMAND                  CREATED    STATUS    PORTS    NAMES
fb86a7697a02   ubuntu:22.04   "/bin/bash"            2 weeks ago   Exited (0) 2 weeks ago           Web
ubuntu@ip-172-31-25-6:~$
```

docker images

```
ubuntu@ip-172-31-25-6:~$ docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
ubuntu        22.04    b1dc6972547a   7 weeks ago   77.9MB
ubuntu@ip-172-31-25-6:~$
```

```
ubuntu@ip-172-31-25-61:~$ docker exec -it Web /bash
OCI runtime exec failed: exec failed: unable to start container process: exec: "/bash": stat /bash: no such file or directory: unknown
ubuntu@ip-172-31-25-61:~$ docker exec -it Web bash
root@fb86a7697a02:/# service apache2 status
 * apache2 is not running
root@fb86a7697a02:/# service apache2 start
 * Starting Apache httpd web server apache2
AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 172.17.0.2. Set the 'ServerName' directive globally to suppress this message
 *
root@fb86a7697a02:/# echo "<h1>Antonio Mateos Delgado</h1>" > /var/www/html/index.html
root@fb86a7697a02:/# service apache2 status
 * apache2 is running
root@fb86a7697a02:/#
```